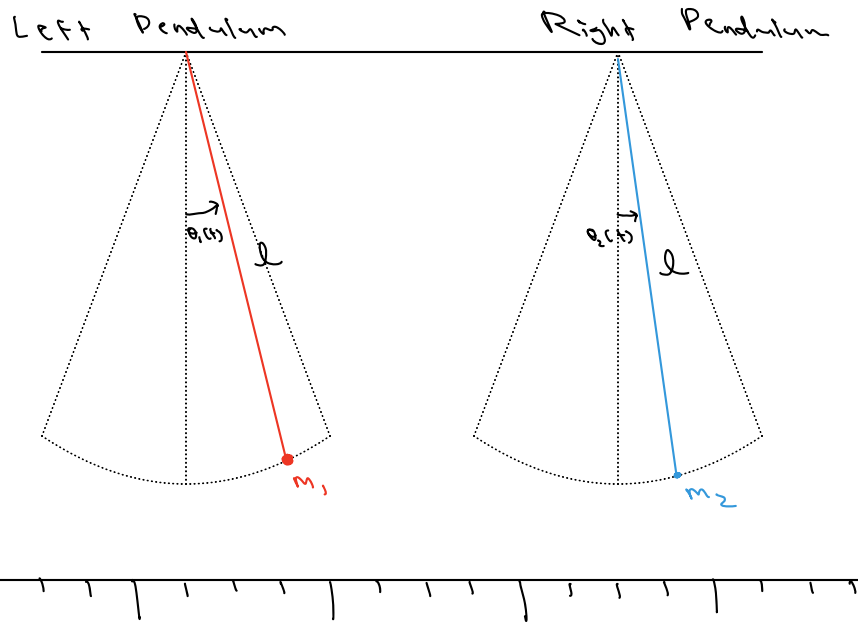


Mathematical Model for Coupled Pendula Problem



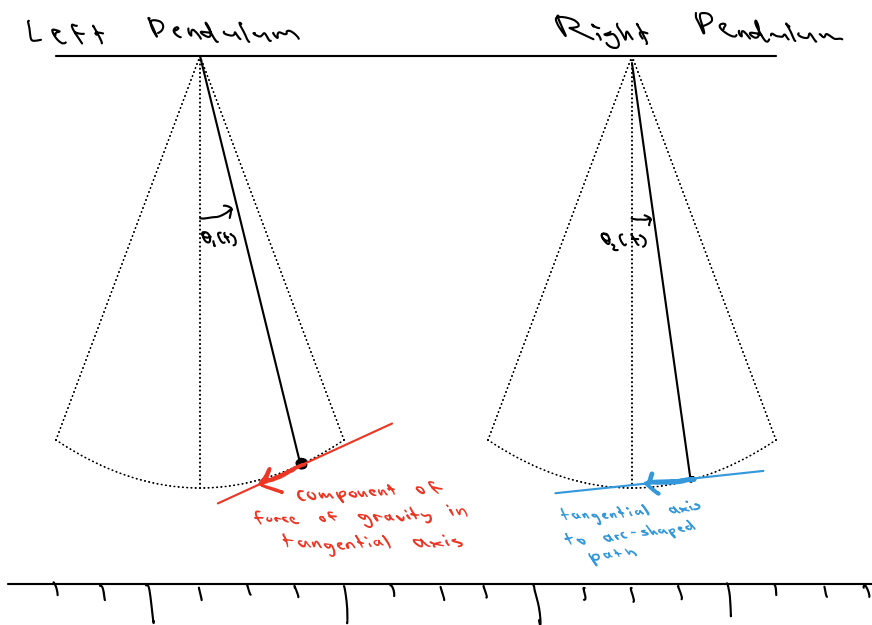
We assume:

- no energy loss
- pendula swing back & forth in the same plane
- enough room so masses don't collide
- same cord length

m_1 = measured mass of left value (in kg)

m_2 = measured mass of right value (in kg)

With these constructions, we have enough to state a 2nd order differential equation for the motion of each pendulum:



Left: $-m_1 \cdot g \cdot \sin(\theta_1(t)) = m_1 \cdot l \cdot \ddot{\theta}_1(t)$

Newton's 2nd Law: $\underbrace{\sum \vec{F}}_{\substack{\text{net force} \\ \text{acting on point} \\ \text{mass}}} = \underbrace{m}_{\substack{\text{mass (kg)}}} \cdot \underbrace{a(t)}_{\substack{\text{acceleration} \\ \text{of point mass} \\ \text{m/s}^2}}$

Right: $-m_2 \cdot g \cdot \sin(\theta_2(t)) = m_2 \cdot l \cdot \ddot{\theta}_2(t)$

We have a system of uncoupled ODE's to describe motion of each pendulum in terms of angle $\theta_i(t)$.

Left: $\ddot{\theta}_1(t) + \frac{g}{l} \sin(\theta_1(t)) = 0$

single var: ordinary
↓

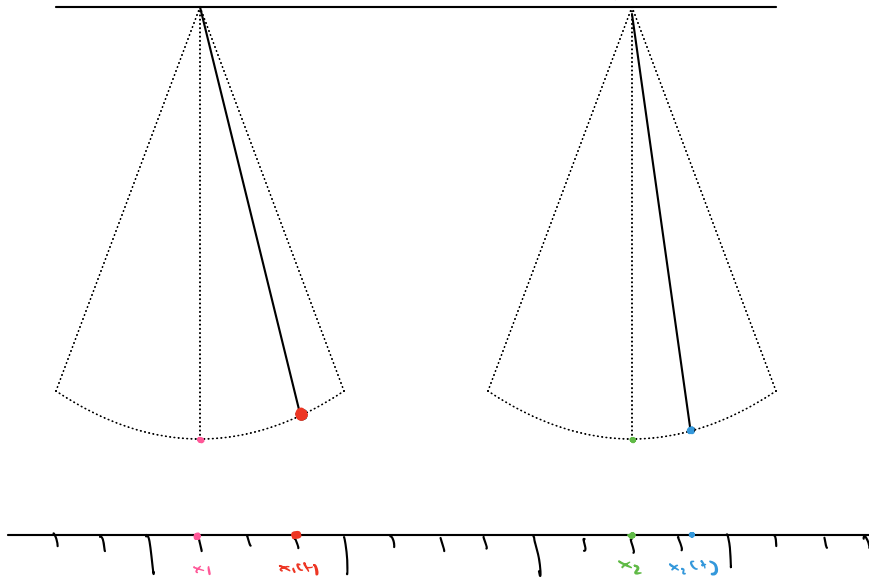
2nd order term nonlinear term
↓ ↓

Right: $\ddot{\theta}_2(t) + \frac{g}{l} \sin(\theta_2(t)) = 0$

Recall that solving these using tools from just Calculus is very difficult. Hence, we **linearize** the equations.

Left Pendulum

Right Pendulum



• shadow on ruler is orthogonal projection

Using our small angle approximation scheme we previously derived, we see

Left:

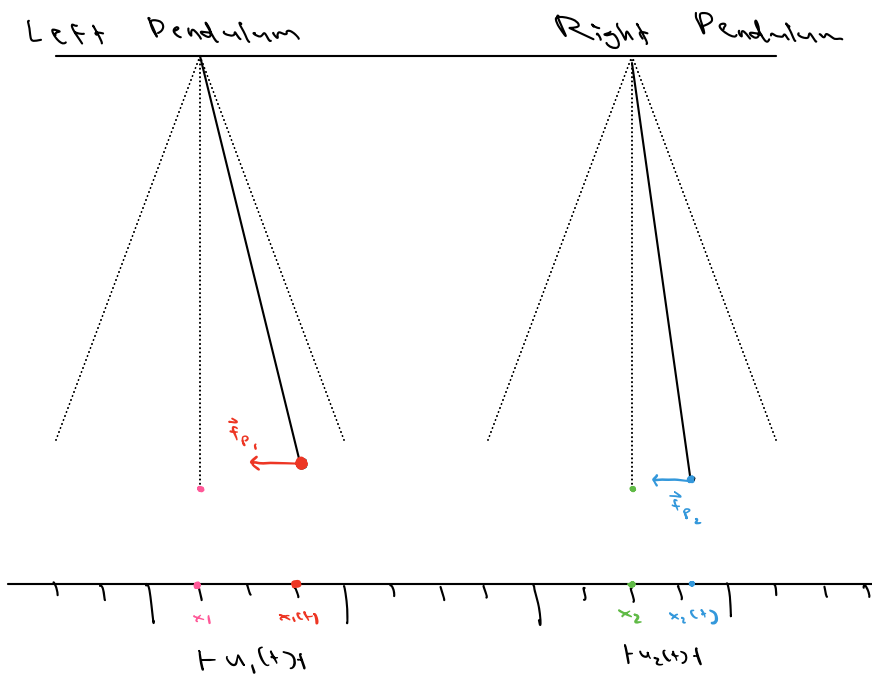
$$\frac{-m_1 \cdot g \cdot u_1(t)}{l} = m_1 \cdot \ddot{u}_1(t)$$

$$\underbrace{\sum \vec{F}}_{\substack{\text{net force} \\ \text{acting on point} \\ \text{mass}}} = \underbrace{m}_{\substack{\text{mass (kg)}}} \cdot \underbrace{a(t)}_{\substack{\text{acceleration} \\ \text{of point mass} \\ \text{m/s}^2}}$$

Right:

$$\frac{-m_2 \cdot g \cdot u_2(t)}{l} = m_2 \cdot \ddot{u}_2(t)$$

Now, we don't need to think about height of bob.



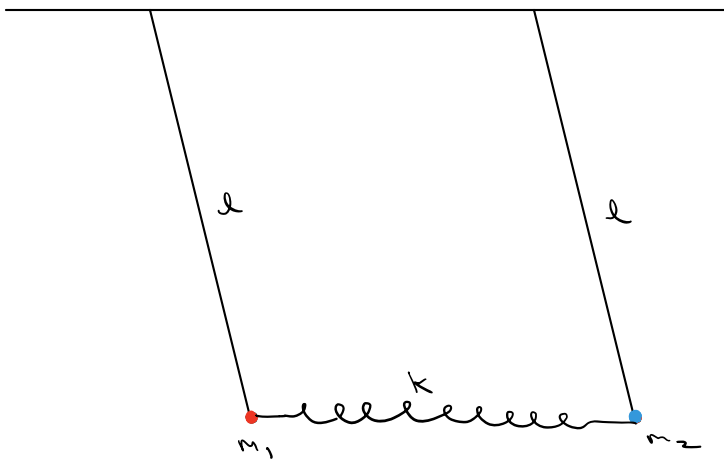
We call the linearized pendulum forces acting in the u -axes approximating the effects of gravity on each bob:

$$\vec{f}_{P_1} = - \frac{m_1 \cdot g}{l} \cdot u_1(t)$$

Forces act in opposite direction of measured displacement

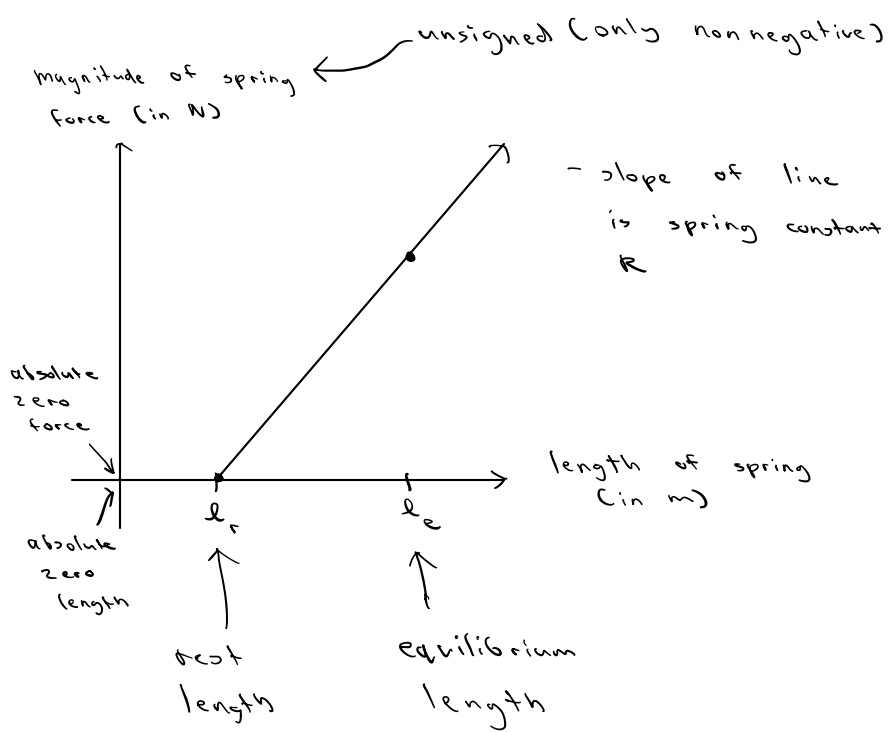
$$\vec{f}_{P_2} = - \frac{m_2 \cdot g}{l} \cdot u_2(t)$$

Now that we know behavior of individual springs, let's couple the masses, using a spring.



We assume:

- spring has zero mass
- when system at rest with spring attached, both masses rest in equilibrium position.
- we measure changes in spring's internal force with respect to the nonzero reference force stored when the spring is at equilibrium
- length of spring at rest is l_r
- length of spring at equilibrium is l_e
where $l_e = x_2 - x_1$
- $l_e - l_r =$ change in length from detached length
- magnitude of internal spring force in equilibrium is $k \cdot (l_e - l_r)$. This is nonzero



Recall Hooke's Law:

change in internal stored force

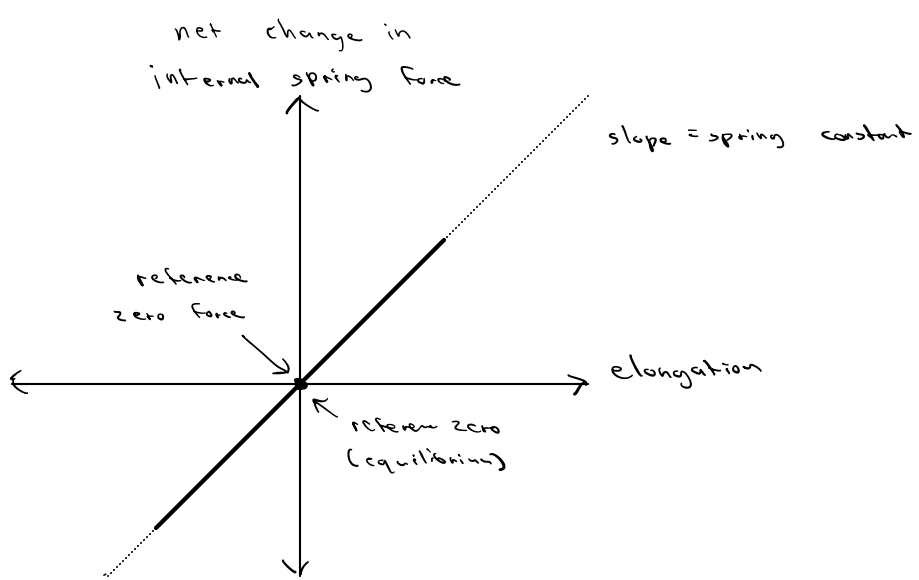
elongation length from some reference

$$F_s = \underbrace{R}_{\text{spring constant}} \cdot \underbrace{e(t)}_{\text{elongation length from some reference}}$$

$$\begin{aligned} \Rightarrow e(t) &= \underbrace{x_2(t) - x_1(t)}_{\text{spring length at time } t} - l_e \\ &= x_2(t) - x_1(t) - (x_2 - x_1) \\ &= x_2(t) - x_2 - x_1(t) + x_1 \\ &= u_2(t) - (x_1(t) - x_1) \\ &= u_2(t) - u_1(t) \end{aligned}$$

$$\begin{aligned} \Rightarrow F_s &= R \cdot e(t) \\ &= R \cdot (u_2(t) - u_1(t)) \end{aligned}$$

- Force positive when masses further apart than at equilibrium.
- Force negative when masses closer together than at equilibrium.



1 h 14 m